

Microeconomics Group Assignment 3

Fogel Study Group 1

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Q38

$$\begin{aligned}
 a) \quad x_1 &= 100K - 10K = 90K & p_1 &= 0.25 \\
 x_2 &= 100K & p_2 &= 0.5 \\
 x_3 &= 100K - 10K = 90K & p_3 &= 0.25 \\
 E(\tilde{x}) &= 0.25(90K) + 0.5(100K) + 0.25(90K) \\
 &= 100,000
 \end{aligned}$$

38b)

If Faruk takes on the project:

$$E[u(\tilde{x})] = \frac{1}{4}(30) + \frac{1}{2}(26) + \frac{1}{4}(20) = 25.5$$

if Faruk does not take project,

$$u(100,000) = 26$$

Therefore, Faruk will not take the project.

$$38c) \text{ Since } E[u(\text{taking project})] = \underbrace{u(99,000)}_{CE} \because \text{both are } 25.5$$

$$\begin{aligned}
 p &= E(\tilde{x}) - CE \\
 &= 100000 - 99000 \\
 &= 1000
 \end{aligned}$$

Risk neutral

39 a) Good outcome = \$400 $\phi_G = 0.5$

Bad outcome = -\$200 $1 - \phi_G$

$$E(\tilde{x}) = 0.5 \times 400 + 0.5(-200) \\ = \$100$$

b) Indifferent to project, set $E(\tilde{x}) = 0$

$$\phi_G(400) + (1 - \phi_G)(-200) = 0$$

$$400\phi_G - 200 + 200\phi_G = 0$$

$$\phi_G = \frac{200}{600} = 0.333$$

c) set bad outcome = -\$100
set $E(\tilde{x}) = 0$ (for indifference/Risk neutral)

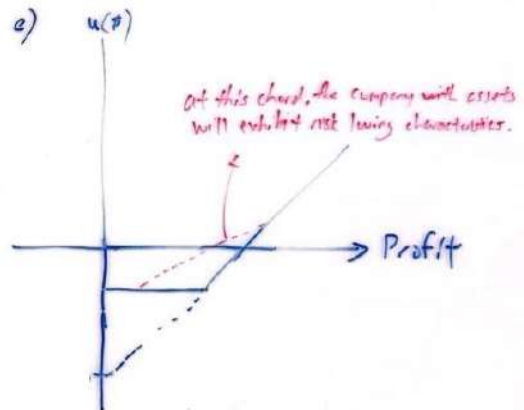
$$\phi_G(400) + (1 - \phi_G)(-100) = 0$$

$$400\phi_G - 100 + 100\phi_G = 0$$

$$\phi_G = \frac{100}{500} = 0.2$$

1) $\phi_G = 0.2$ (from part c)

$$E(\tilde{x}) = \cancel{0.2(400)} + \cancel{0.8(0)} \neq 80 \\ 0.2(400) + 0.8(0) = \$80$$



Q43

$$X_1 \quad 2000 \quad \phi 0.5 \quad -$$

$$X_2 \quad 1000 \quad \phi 0.25$$

$$X_3 \quad -5000 \quad \phi 0.25$$

$$\begin{aligned} E(\tilde{X}) &= 0.5(2000) + 0.25(1000) + 0.25(-5000) \\ &= 750 \end{aligned}$$

A risk averse person will always prefer the ST (sure thing) outcome to any risky portfolio with the same outcome.

Therefore, this person is risk-averse. True